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1.0 Cloud Infrastructure Overview & Microservices Architecture
The production platform is hosted on AWS with an active-active multi-region topology spanning us-east-1 (N. Virginia) and eu-central-1 (Frankfurt). All microservices are containerized via Docker and orchestrated with Amazon EKS (Elastic Kubernetes Service). Each microservice maintains dedicated ingress controllers, Envoy service mesh proxies, and isolated IAM roles.

2.0 High-Throughput Caching Tier
Data caching utilizes an Amazon ElastiCache Redis cluster version 7.1 with 16 distributed shards. Cluster settings enforce a maxmemory allocation of 32 GB per node with allkeys-lru eviction policy. Target cache hit ratios must maintain above 94.5% to protect underlying Aurora PostgreSQL database read replicas.

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3.0 Canary Deployments & CI/CD Pipeline

Continuous delivery is driven via GitHub Actions and ArgoCD. Every pull request passes automated unit tests, SonarQube static security analysis, and container vulnerability scanning with Trivy. Production rollouts employ progressive canary deployments:

- Stage 1: 5% traffic routing for 15 minutes with automated error rate monitoring.
- Stage 2: 25% traffic routing for 30 minutes.
- Stage 3: 100% full cutover if p99 latency remains under 120ms and HTTP 5xx errors are below 0.01%.

4.0 Disaster Recovery & Recovery Point Objectives

Automated cross-region database snapshots execute every 4 hours. The Recovery Point Objective (RPO) is strictly designated at 15 minutes, while the Recovery Time Objective (RTO) is guaranteed at 45 minutes for full regional failover.